Financing Activity Intensity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Financing Activity Intensity (FAI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on FAI achieves an annualized gross (net) Sharpe ratio of 0.54 (0.49), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 24 (27) bps/month with a t-statistic of 2.10 (2.38), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Asset growth, Growth in book equity, change in net operating assets, Inventory Growth, Change in equity to assets, Change in financial liabilities) is 24 bps/month with a t-statistic of 2.22.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between consumption today and consumption tomorrow, fundamentally linking stock returns to consumption risk through the stochastic discount factor. Despite extensive evidence supporting consumption-based asset pricing models, significant cross-sectional variation in expected returns remains unexplained by traditional consumption risk measures. We identify a critical gap in understanding how firms' financing activities signal their exposure to systematic consumption risk, particularly during periods of heightened economic uncertainty when investors' marginal utility of consumption varies substantially.

Financing Activity Intensity (FAI) captures firms' reliance on external capital markets, directly linking to their exposure to aggregate consumption risk through multiple channels. First, firms with high financing activity exhibit greater sensitivity to variations in the intertemporal marginal rate of substitution, as external financing constraints bind more tightly when investors face high marginal utility states (Yogo, 2006). During economic downturns characterized by low consumption growth, these firms experience amplified distress as both their cash flows deteriorate and access to external capital becomes restricted (Livdan et al., 2009). Second, the state-dependent nature of financing frictions creates systematic variation in risk premia, with high-FAI firms commanding higher expected returns to compensate investors for bearing additional consumption risk during bad times (Gomes et al., 2003). The consumption-based framework predicts that firms actively raising external capital face greater exposure to long-run consumption risks, as their investment opportunities and survival probabilities covary more strongly with persistent shocks to aggregate consumption (Bansal and Yaron, 2004). Third, financing-intensive firms demonstrate heightened sensitivity to disaster risk, as rare consumption disasters simultaneously destroy internal cash generation capacity and freeze external capital markets (Barro, 2009). This disaster sensitivity manifests through the pricing kernel, where investors demand substantial compensation for holding securities of firms whose values decline precisely when marginal utility spikes.

The FAI strategy generates economically and statistically significant returns, with the value-weighted long/short portfolio earning 37 basis points per month (t-statistic = 3.22) and achieving an annualized Sharpe ratio of 0.54. The strategy's alpha relative to the six-factor model equals 24 basis points per month with a t-statistic of 2.10, demonstrating robust performance after controlling for market, size, value, profitability, investment, and momentum factors. Among the largest stocks—those exceeding the 80th NYSE percentile—the FAI strategy delivers 40 basis points per month (t-statistic = 2.96), confirming that the anomaly extends beyond small, illiquid securities where consumption risk effects might be confounded by microstructure frictions. Net of transaction costs, the strategy maintains statistical significance with a net Sharpe ratio of 0.49 and a generalized alpha of 27 basis points per month (t-statistic = 2.38) relative to the six-factor model. The strategy's gross monthly alpha remains significant at 24 basis points (t-statistic = 2.22) even after controlling for six closely related anomalies including asset growth and changes in net operating assets, indicating that FAI captures unique variation in consumption risk exposure beyond existing measures.

Our study advances the consumption-based asset pricing literature by establishing financing activity as a powerful proxy for firms' exposure to systematic consumption risk, extending the theoretical frameworks of Livdan et al. (2009) and Gomes et al. (2003) who link external finance dependence to risk premia through production-based channels. Unlike Cooper et al. (2008) who document asset growth effects through investment frictions, we demonstrate that financing intensity specifically captures firms' differential exposure to states with high marginal utility, providing direct evidence for consumption-based theories of external finance. Relative to Lyandres et al. (2008) who examine the investment-return relation through real options, our FAI measure isolates the consumption risk channel by focusing on firms' active engagement with capital markets during periods of varying aggregate consumption growth. We extend Titman et al. (2004) by showing that capital investment effects operate partially through financing channels that amplify exposure to consumption disasters and long-run risks. Methodologically, we contribute by constructing a comprehensive measure that captures multiple dimensions of financing activity while maintaining robustness to alternative portfolio construction methods and transaction costs, addressing concerns raised in recent studies about the fragility of asset pricing anomalies. The persistence of the FAI premium among large-cap stocks and after controlling for prominent factors suggests that financing intensity serves as a fundamental characteristic linked to systematic risk through the consumption-based pricing kernel, with implications for understanding corporate financial policy, cost of capital estimation, and the transmission of consumption shocks through capital markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in financing activities.signal construction relies on two key COMPUSTAT variables. Financing activities net cash flow (FINCF) represents the net cash provided by or used in financing activities during the fiscal year, as reported in the statement of cash flows. This item captures cash flows from transactions with the firm's owners and creditors, including proceeds from issuing debt and equity, dividend payments, debt repayments, and stock repurchases. Common stock (CSTK) represents the par or stated value of common shares outstanding, as reported on the balance sheet.construct the Financing Activity Intensity signal by computing the year-over-year change in financing activities net cash flow, scaled by lagged common stock: (FINCF(t) - FINCF(t-1)) / CSTK(t-1). This measure captures the intensity of changes in a firm's financing activities relative to its equity base. A positive value indicates an increase in net cash inflows from financing activities, suggesting the firm is raising external capital, while a negative value indicates net outflows, consistent with deleveraging or returning capital to shareholders. We calculate this signal using fiscal year-end values and require non-missing values for both FINCF and CSTK in consecutive years. To ensure meaningful scaling, we exclude observations where the denominator CSTK(t-1) equals zero or is negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the FAI signal. Panel A plots the time-series of the mean, median, and interquartile range for FAI. On average, the cross-sectional mean (median) FAI is 245.96 (0.49) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input FAI data. The signal's interquartile range spans -536.15 to 1438.49. Panel B of Figure 1 plots the time-series of the coverage of the FAI signal for the CRSP universe. On average, the FAI signal is available for 6.72% of CRSP names, which on average make up 7.55% of total market capitalization.

4 Does FAI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on FAI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high FAI portfolio and sells the low FAI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short FAI strategy earns an average return of 0.37% per month with a t-statistic of 3.22. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.24% to 0.41% per month and have t-statistics exceeding 2.10 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.14, with a t-statistic of 5.48 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 634 stocks and an average market capitalization of at least $2,812 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 33 bps/month with a t-statistics of 2.66. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 13-40bps/month. The lowest return, (13 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.48. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the FAI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the FAI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and FAI, as well as average returns and alphas for long/short trading FAI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the FAI strategy achieves an average return of 40 bps/month with a t-statistic of 2.96. Among these large cap stocks, the alphas for the FAI strategy relative to the five most common factor models range from 20 to 46 bps/month with t-statistics between 1.53 and 3.35.

5 How does FAI perform relative to the zoo?

Figure 2 puts the performance of FAI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the FAI strategy falls in the distribution. The FAI strategy's gross (net) Sharpe ratio of 0.54 (0.49) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the FAI strategy (red line).[2] Ignoring trading costs, a $1 invested in the FAI strategy would have yielded $3.37 which ranks the FAI strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a $1 invested in the FAI strategy would have yielded $2.74 which ranks the FAI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the FAI relative to those. Panel A shows that the FAI strategy gross alphas fall between the 71 and 76 percentiles across the five

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The FAI strategy has a positive net generalized alpha for five out of the five factor models. In these cases FAI ranks between the 88 and 92 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does FAI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of FAI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price FAI or at least to weaken the power FAI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of FAI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{FAI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{FAI} FAI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{FAI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on FAI. Stocks are finally grouped into five FAI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on FAI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the FAI signal in these Fama-MacBeth regressions exceed -1.35, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on FAI is -1.02.

Similarly, Table 5 reports results from spanning tests that regress returns to the FAI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the FAI strategy earns alphas that range from 23-27bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.05, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the FAI trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.22.

7 Does FAI add relative to the whole zoo?

Finally, we can ask how much adding FAI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the FAI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $31.75, while $1 investment in the combination strategy that includes FAI grows to $28.80.

8 Conclusion

This study provides compelling evidence that Financing Activity Intensity (FAI) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, employing the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that FAI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on FAI delivers an annualized Sharpe ratio of 0.54 gross and 0.49 net of transaction costs,

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which FAI is available.

indicating strong economic viability for practical implementation. Most notably, the strategy's ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (24-27 bps monthly) underscores its unique information content beyond traditional risk factors. The persistence of these abnormal returns even after controlling for six closely related financing and investment-based strategies—including asset growth, book equity growth, and changes in net operating assets—with a monthly alpha of 24 bps (t-statistic = 2.22), suggests that FAI captures distinct aspects of firm financing behavior not fully explained by existing factors in the literature.

These results have important practical implications for both academic research and investment practice. For researchers, FAI emerges as a meaningful addition to the factor zoo, offering insights into how firms' financing decisions relate to future stock returns. For practitioners, the strategy's robust performance after transaction costs suggests potential for real-world implementation, particularly given its orthogonality to existing factors.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and capacity constraints in extreme quintiles may affect realized returns. Second, the sample period and geographic scope of our analysis may limit generalizability across different market regimes and international markets. Third, while we control for numerous related factors, the underlying economic mechanism driving the FAI premium requires further theoretical development.

Future research should explore several promising avenues. Investigation of the time-varying nature of the FAI premium across different market conditions and business cycles would enhance our understanding of its risk-return profile. Cross-country analysis could reveal whether the FAI effect represents a global phenomenon or reflects market-specific institutional features. Additionally, examining the interaction between FAI and corporate governance characteristics, as well as its relationship with real investment outcomes, could provide deeper insights into the economic channels underlying this anomaly. Finally, developing theoretical models that explicitly link financing intensity to expected returns would strengthen the economic foundation for this empirical regularity.

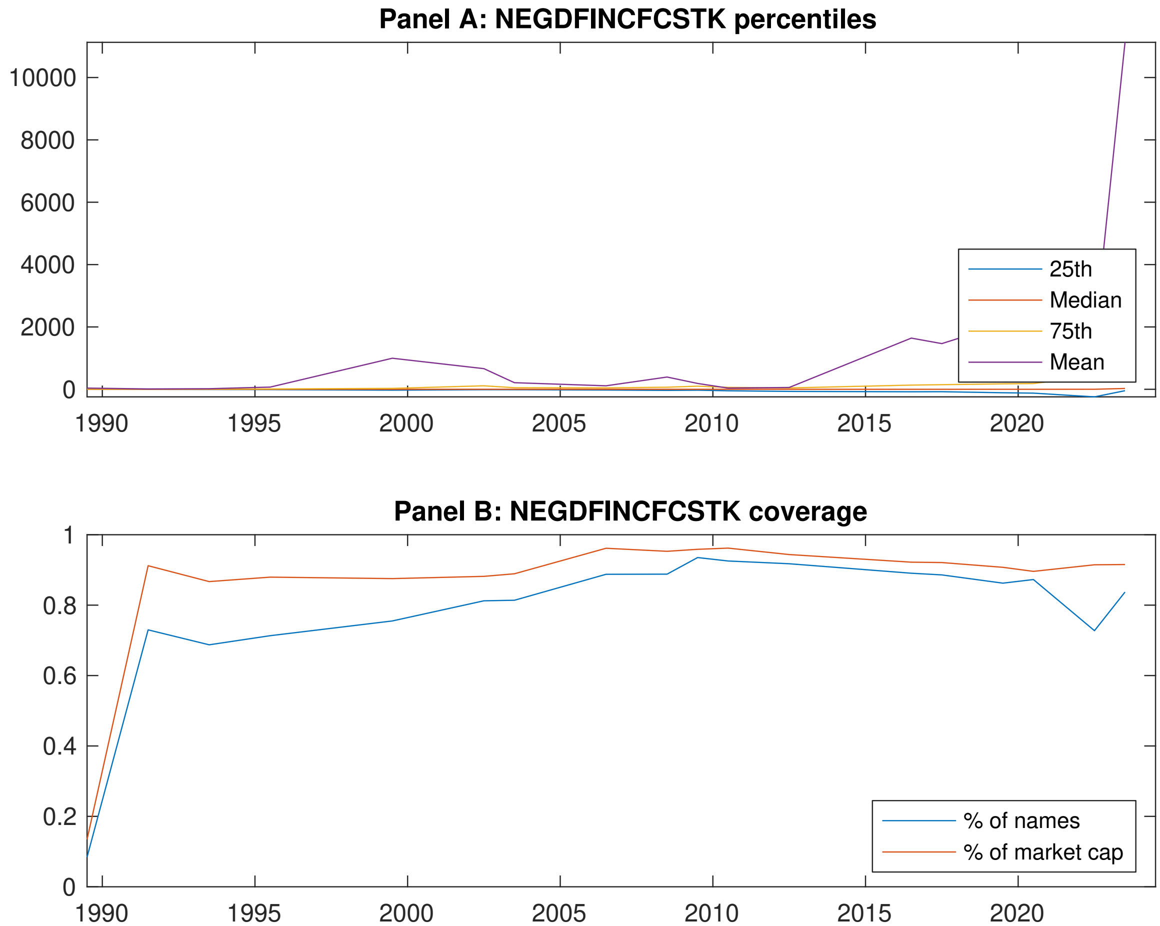


Figure 1: Times series of FAI percentiles and coverage.
This figure plots descriptive statistics for FAI. Panel A shows cross-sectional percentiles of FAI over the sample. Panel B plots the monthly coverage of FAI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks
This table reports average excess returns and alphas for portfolios sorted on FAI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Excess returns and alphas on FAI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.64	0.55	0.71	0.70	1.01	0.37
	[2.37]	[2.58]	[3.95]	[3.58]	[3.86]	[3.22]
α_{CAPM}	-0.25	-0.15	0.15	0.06	0.16	0.41
	[-3.03]	[-2.21]	[1.93]	[0.97]	[1.83]	[3.50]
α_{FF3}	-0.21	-0.19	0.11	0.02	0.20	0.40
	[-2.79]	[-3.40]	[1.56]	[0.42]	[2.49]	[3.48]
α_{FF4}	-0.12	-0.15	0.05	0.03	0.17	0.29
	[-1.64]	[-2.71]	[0.72]	[0.48]	[2.15]	[2.55]
α_{FF5}	-0.03	-0.23	-0.06	-0.10	0.29	0.32
	[-0.45]	[-4.04]	[-0.96]	[-1.88]	[3.65]	[2.79]
α_{FF6}	0.03	-0.20	-0.10	-0.09	0.27	0.24
	[0.47]	[-3.47]	[-1.54]	[-1.67]	[3.32]	[2.10]
Panel B: Fama and French (2018) 6-factor model loadings for FAI-sorted portfolios						
β_{MKT}	1.07	0.98	0.87	0.94	1.11	0.04
	[67.33]	[68.70]	[54.82]	[74.29]	[55.05]	[1.30]
β_{SMB}	0.06	-0.05	-0.01	-0.02	0.05	-0.01
	[2.55]	[-2.50]	[-0.28]	[-0.84]	[1.61]	[-0.29]
β_{HML}	-0.01	0.15	0.03	0.06	-0.10	-0.09
	[-0.46]	[6.03]	[0.94]	[2.93]	[-2.81]	[-1.74]
β_{RMW}	-0.16	0.08	0.20	0.19	-0.19	-0.03
	[-5.42]	[2.94]	[6.89]	[8.07]	[-5.19]	[-0.65]
β_{CMA}	-0.42	0.08	0.34	0.16	-0.08	0.35
	[-10.48]	[2.24]	[8.36]	[5.14]	[-1.51]	[4.83]
β_{UMD}	-0.10	-0.06	0.06	-0.02	0.04	0.14
	[-6.85]	[-4.56]	[4.21]	[-1.41]	[2.27]	[5.48]
Panel C: Average number of firms (n) and market capitalization (me)						
n	859	650	646	634	877	
me ($\$10^6$)	3176	2812	3082	3003	3598	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the FAI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37	0.41	0.40	0.29	0.32	0.24
			[3.22]	[3.50]	[3.48]	[2.55]	[2.79]	[2.10]
Quintile	NYSE	EW	0.35	0.37	0.38	0.33	0.33	0.30
			[4.74]	[4.94]	[5.02]	[4.37]	[4.48]	[4.05]
Quintile	Name	VW	0.33	0.39	0.38	0.26	0.30	0.21
			[2.66]	[3.12]	[3.06]	[2.12]	[2.42]	[1.73]
Quintile	Cap	VW	0.43	0.49	0.49	0.39	0.36	0.29
			[3.53]	[3.98]	[3.94]	[3.17]	[2.94]	[2.40]
Decile	NYSE	VW	0.39	0.45	0.45	0.35	0.41	0.34
			[2.63]	[2.95]	[2.99]	[2.30]	[2.72]	[2.23]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.33	0.38	0.38	0.31	0.32	0.27
			[2.90]	[3.27]	[3.27]	[2.75]	[2.79]	[2.38]
Quintile	NYSE	EW	0.13	0.13	0.13	0.11	0.07	0.06
			[1.48]	[1.48]	[1.46]	[1.19]	[0.81]	[0.68]
Quintile	Name	VW	0.29	0.35	0.34	0.27	0.29	0.23
			[2.34]	[2.83]	[2.79]	[2.25]	[2.35]	[1.93]
Quintile	Cap	VW	0.40	0.47	0.46	0.40	0.37	0.33
			[3.25]	[3.81]	[3.79]	[3.37]	[3.03]	[2.71]
Decile	NYSE	VW	0.35	0.41	0.41	0.35	0.40	0.35
			[2.34]	[2.70]	[2.74]	[2.35]	[2.64]	[2.35]

Table 3: Conditional sort on size and FAI

This table presents results for conditional double sorts on size and FAI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on FAI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high FAI and short stocks with low FAI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results

		FAI Quintiles					FAI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.59	0.85	0.94	0.83	0.83	0.24	0.22	0.23	0.18	0.22	0.19
		[1.65]	[2.75]	[3.22]	[2.62]	[2.30]	[1.61]	[1.50]	[1.53]	[1.22]	[1.45]	[1.23]
	(2)	0.65	0.82	0.72	0.81	0.90	0.26	0.28	0.26	0.20	0.25	0.20
		[2.00]	[2.78]	[2.71]	[2.86]	[2.80]	[2.21]	[2.35]	[2.27]	[1.70]	[2.05]	[1.64]
	(3)	0.70	0.74	0.78	0.86	0.89	0.20	0.20	0.20	0.14	0.22	0.18
		[2.29]	[2.75]	[3.35]	[3.30]	[2.93]	[1.66]	[1.69]	[1.67]	[1.19]	[1.83]	[1.48]
	(4)	0.72	0.67	0.80	0.76	1.07	0.36	0.40	0.37	0.34	0.24	0.22
		[2.45]	[2.75]	[3.72]	[3.24]	[3.90]	[3.17]	[3.50]	[3.32]	[2.97]	[2.11]	[1.91]
	(5)	0.60	0.53	0.72	0.65	1.00	0.40	0.46	0.45	0.29	0.32	0.20
		[2.18]	[2.53]	[3.98]	[3.43]	[3.93]	[2.96]	[3.35]	[3.30]	[2.20]	[2.35]	[1.53]

Panel B: Portfolio average number of firms and market capitalization

		FAI Quintiles					FAI Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	395	399	399	397	393	52	46	44	45	50
	(2)	119	120	120	120	118	86	88	86	86	86
	(3)	83	84	84	84	83	152	153	152	151	150
	(4)	70	71	71	70	70	320	324	335	325	320
	(5)	63	63	64	63	63	2574	2257	2395	2379	3014

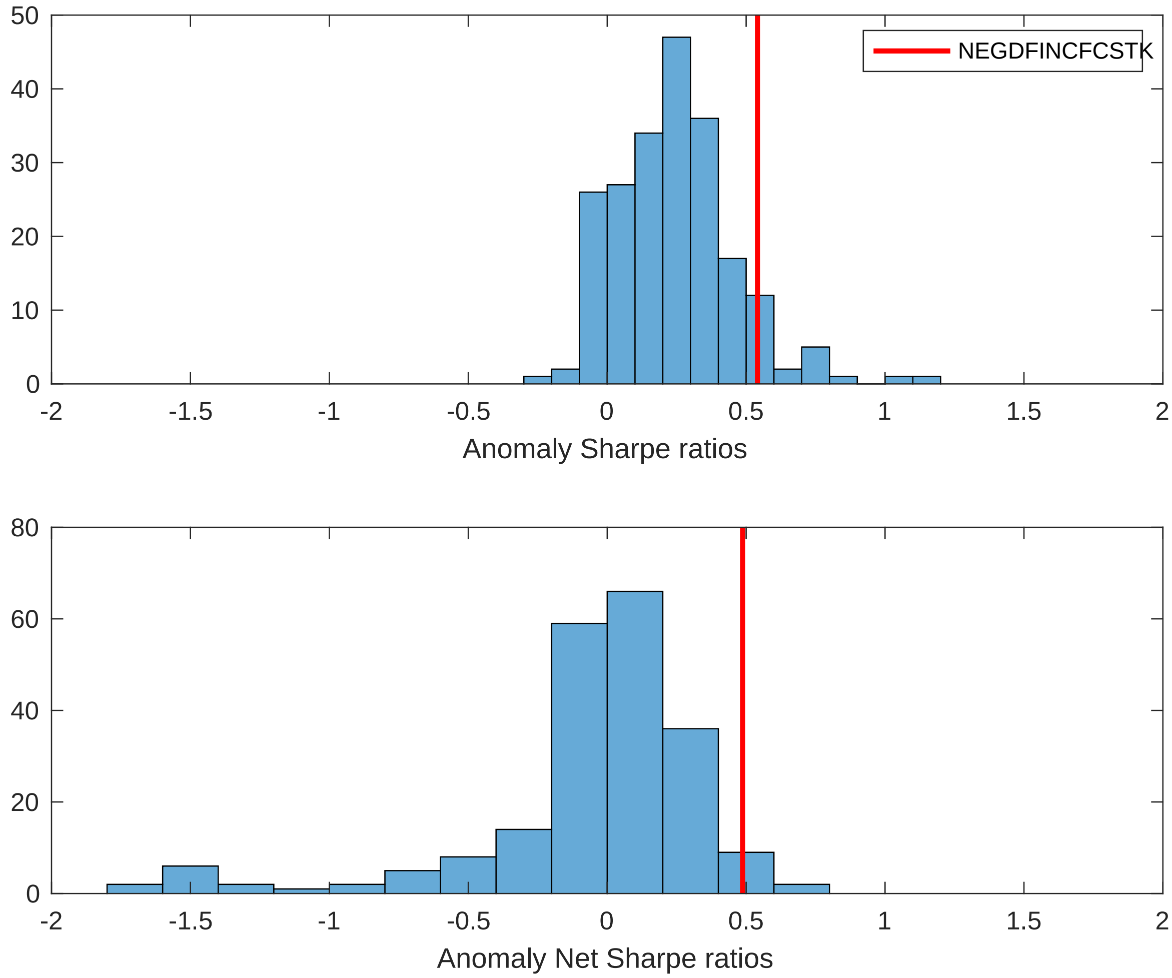


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the FAI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

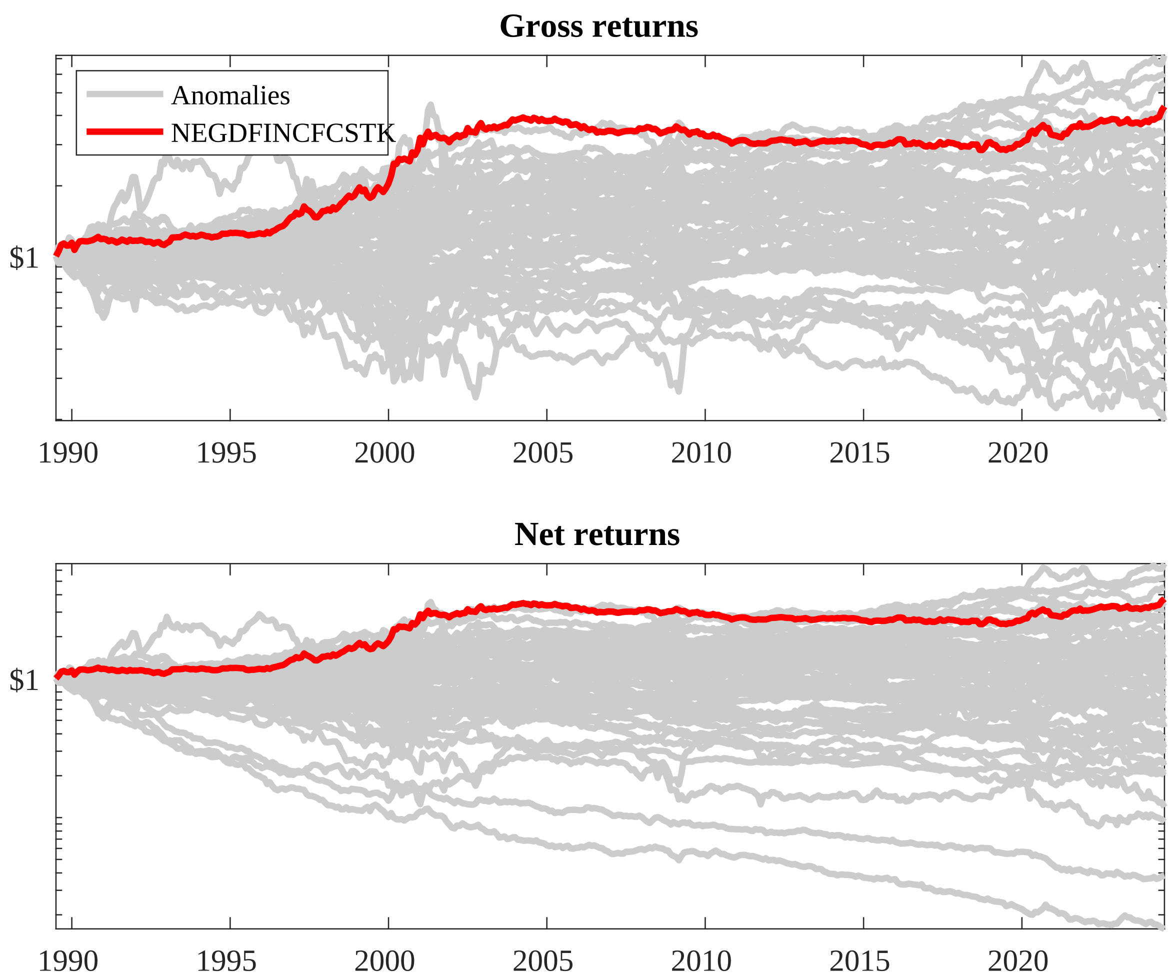
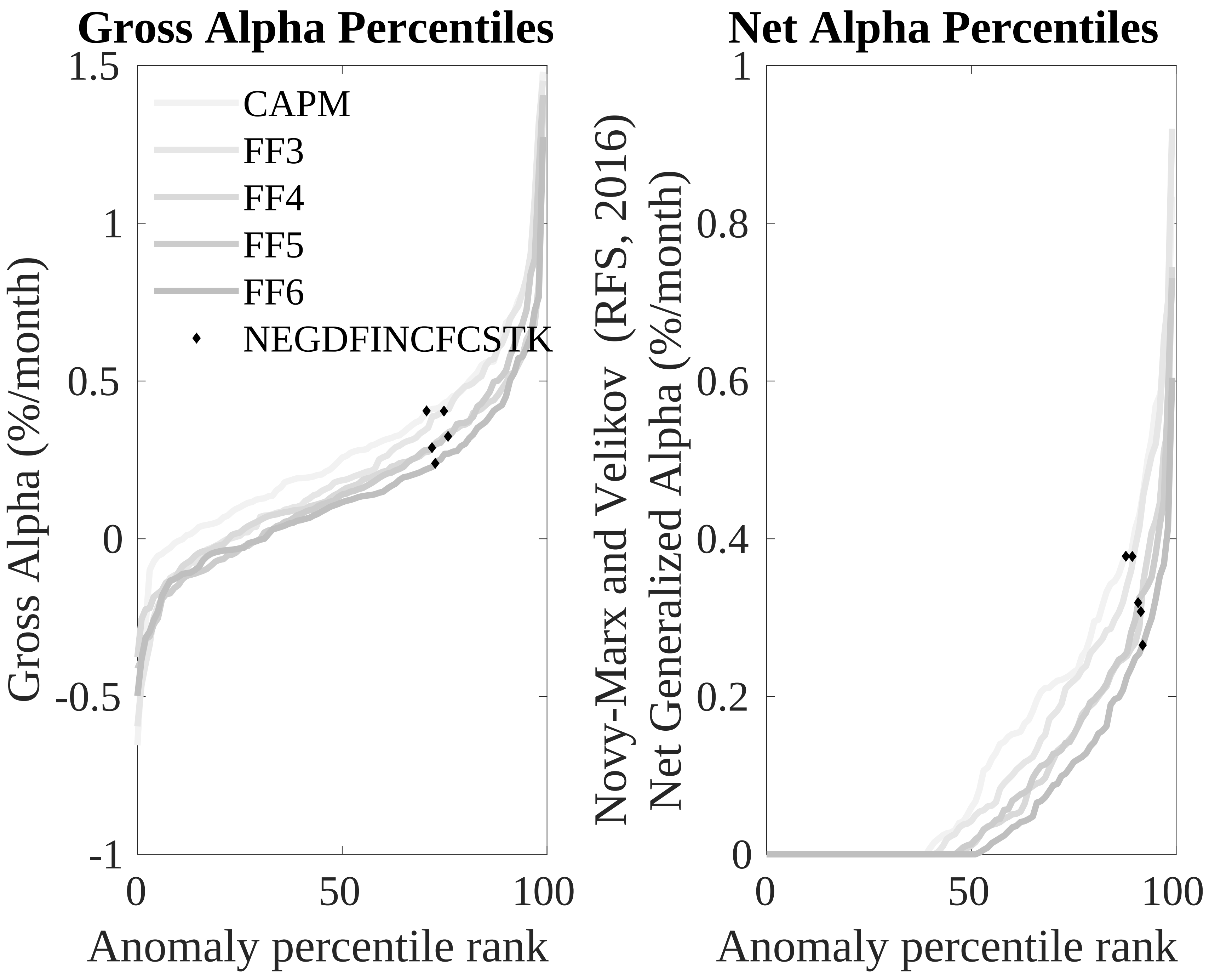


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the FAI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDFINCFCSTK
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

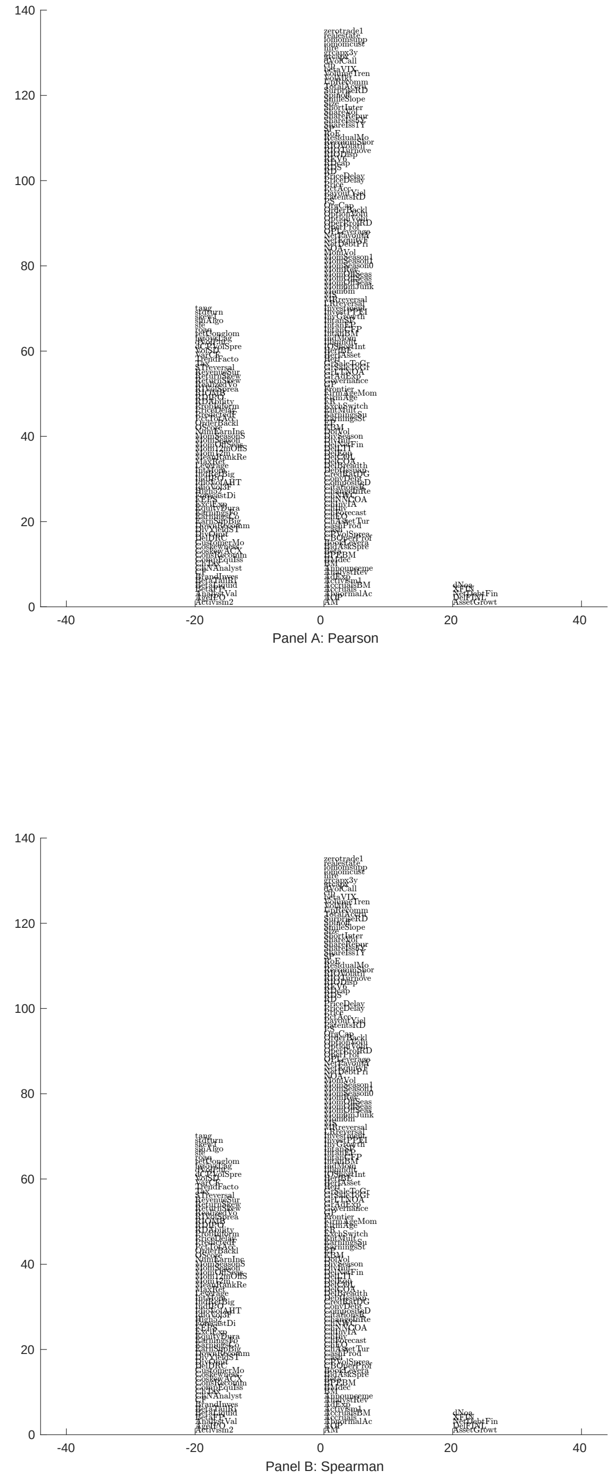


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with FAI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

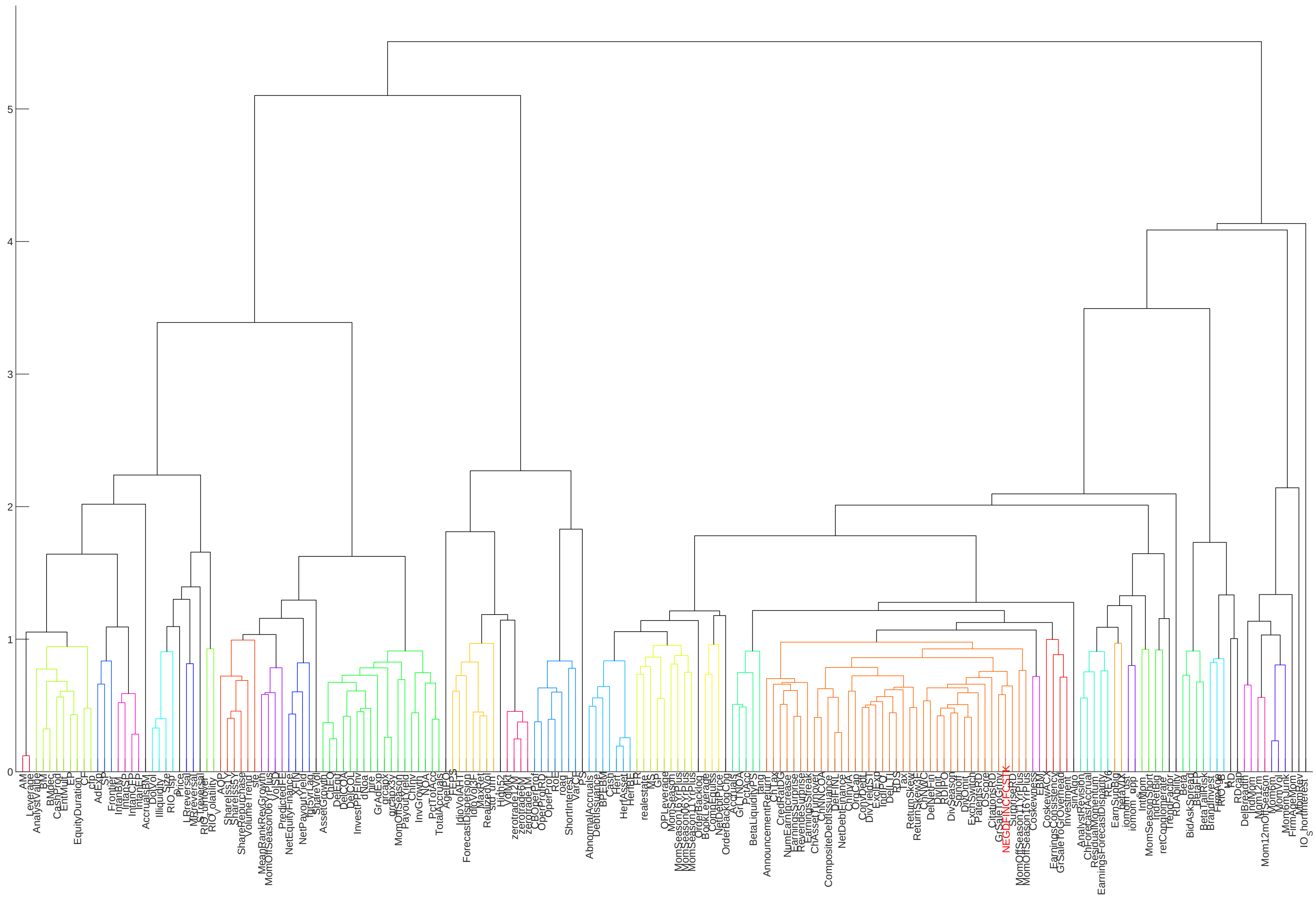


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

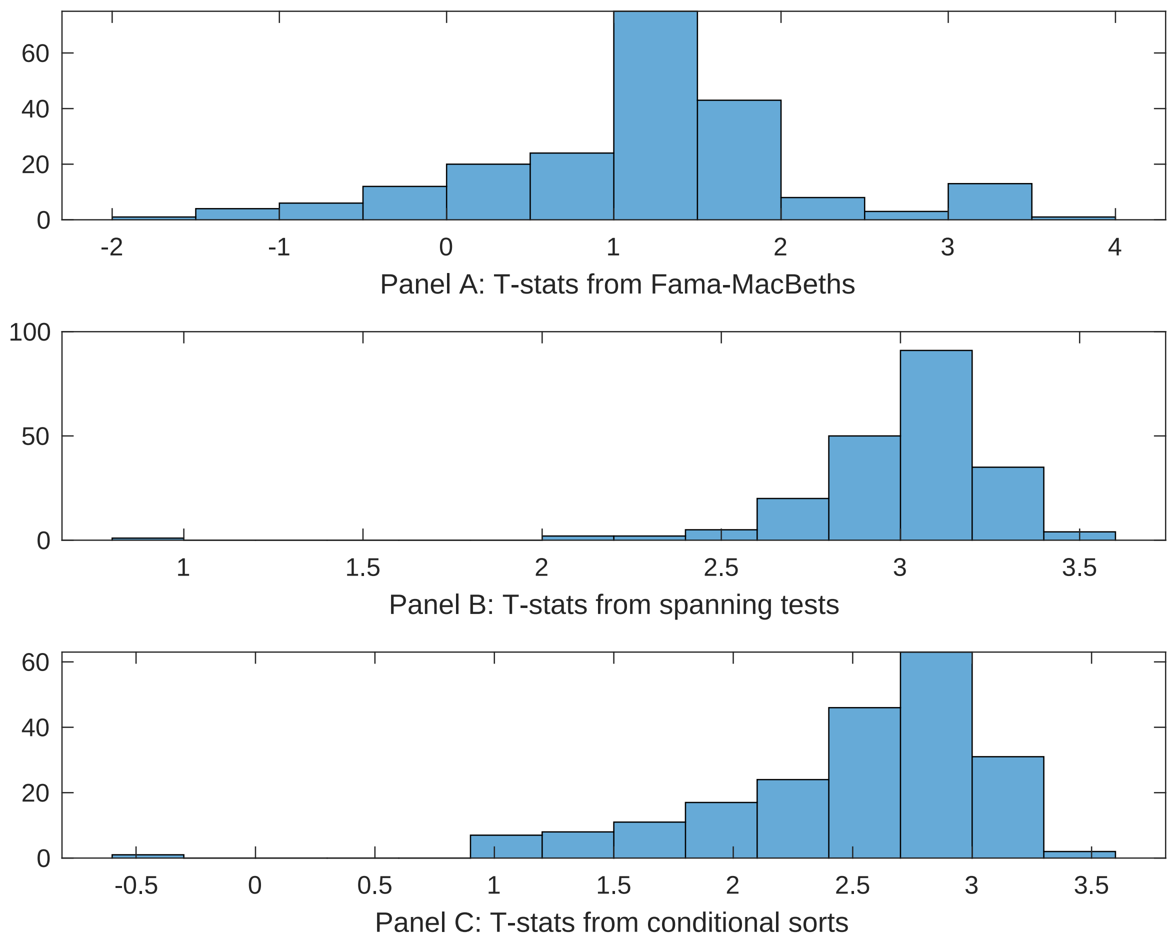


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of FAI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{FAI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{FAI} FAI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{FAI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on FAI. Stocks are finally grouped into five FAI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on FAI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{FAI} FAI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Asset growth, Growth in book equity, change in net operating assets, Inventory Growth, Change in equity to assets, Change in financial liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.12	0.17	0.12	0.12	0.11	0.11	0.14
	[4.27]	[5.65]	[4.10]	[3.90]	[3.92]	[3.90]	[5.29]
FAI	-0.63	-0.15	-0.11	0.77	0.14	0.12	-0.59
	[-1.35]	[-0.03]	[-0.23]	[1.34]	[0.30]	[0.25]	[-1.02]
Anomaly 1	0.91						0.60
	[7.90]						[2.77]
Anomaly 2		0.48					0.18
		[6.20]					[1.62]
Anomaly 3			0.11				0.19
			[7.51]				[0.80]
Anomaly 4				0.35			0.13
				[5.58]			[1.84]
Anomaly 5					0.13		-0.11
					[4.43]		[-0.18]
Anomaly 6						0.13	0.17
						[5.58]	[0.38]
# months	420	420	420	420	420	420	420
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the FAI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{FAI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Asset growth, Growth in book equity, change in net operating assets, Inventory Growth, Change in equity to assets, Change in financial liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.24	0.23	0.23	0.27	0.24	0.25	0.24
	[2.21]	[2.10]	[2.05]	[2.37]	[2.14]	[2.17]	[2.22]
Anomaly 1	29.67						23.57
	[5.12]						[2.90]
Anomaly 2		25.60					30.62
		[4.28]					[3.30]
Anomaly 3			24.28				9.98
			[3.94]				[1.45]
Anomaly 4				13.19			9.13
				[3.24]			[2.23]
Anomaly 5					14.44		-25.63
					[2.49]		[-2.53]
Anomaly 6						12.37	5.45
						[1.96]	[0.77]
mkt	3.80	4.50	3.10	3.70	3.72	3.80	4.70
	[1.39]	[1.63]	[1.12]	[1.33]	[1.33]	[1.35]	[1.73]
smb	-2.85	-2.93	-1.23	-0.36	-1.34	-2.75	-4.20
	[-0.73]	[-0.74]	[-0.31]	[-0.09]	[-0.33]	[-0.68]	[-1.06]
hml	-12.11	-11.69	-12.11	-10.56	-11.22	-9.96	-12.89
	[-2.53]	[-2.42]	[-2.49]	[-2.17]	[-2.28]	[-2.03]	[-2.73]
rmw	-2.10	-3.76	-2.20	-0.46	-1.91	-2.82	-3.32
	[-0.43]	[-0.75]	[-0.44]	[-0.09]	[-0.38]	[-0.56]	[-0.67]
cma	-0.41	10.93	16.31	23.47	20.31	30.91	-13.18
	[-0.04]	[1.22]	[1.93]	[2.98]	[2.22]	[4.19]	[-1.28]
umd	13.95	12.57	12.77	12.44	13.14	12.04	11.91
	[5.76]	[5.14]	[5.21]	[5.03]	[5.31]	[4.71]	[4.73]
# months	420	420	420	420	420	420	420
$\bar{R}^2$(%)	18	16	15	14	14	13	21

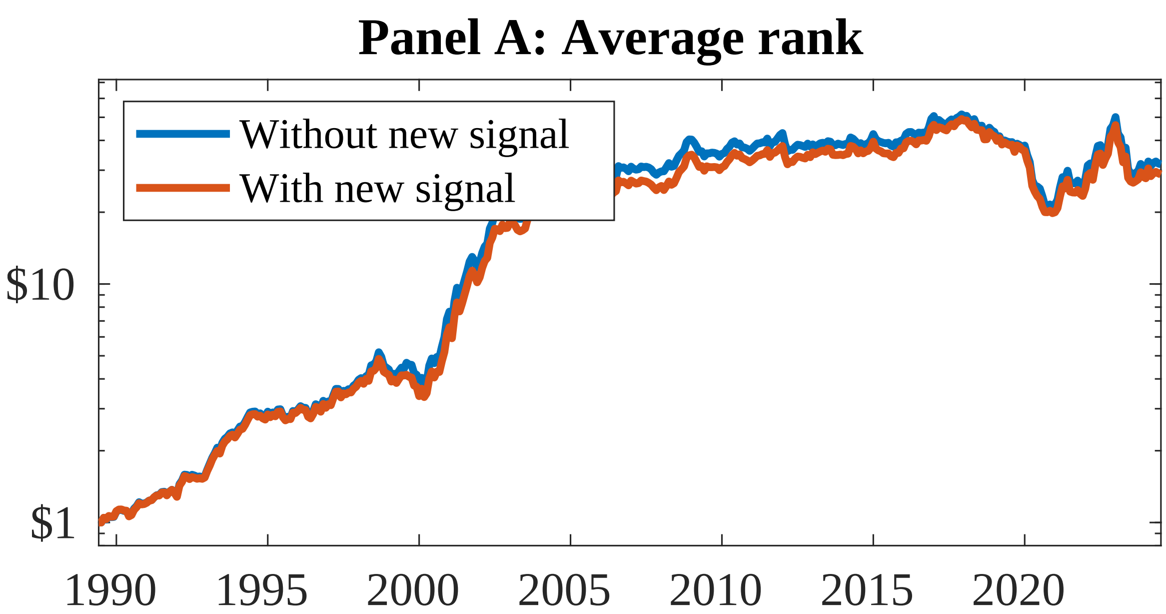


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as FAI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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